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import os
import logging
import requests
from pymongo import MongoClient, UpdateOne
from pymongo.errors import PyMongoError
from dotenv import load_dotenv

logging.basicConfig(
    level=logging.INFO,
    format="%(asctime)s - %(levelname)s - %(message)s"
)

BASE_URL = "https://api.openf1.org/v1"
DB_NAME = "openf1_data"

def conectar_mongodb():
    load_dotenv()
    mongo_uri = os.getenv("MONGO_URI", "mongodb://localhost:27017/")

    try:
        logging.info("Iniciando conexão com o MongoDB...")
        client = MongoClient(mongo_uri, serverSelectionTimeoutMS=5000)
        client.admin.command('ping')
        db = client[DB_NAME]
        logging.info(f"Conectado com sucesso ao banco de dados: {DB_NAME}")
        return db
    except PyMongoError as e:
        logging.error(f"Falha crítica ao conectar no MongoDB: {e}")
        raise SystemExit(e)

def fetch_data(endpoint: str, params: dict) -> list:
    url = f"{BASE_URL}/{endpoint}"
    try:
        logging.info(f"Buscando {endpoint} | Parâmetros: {params}...")
        response = requests.get(url, params=params, timeout=20)
        response.raise_for_status()

        data = response.json()
        logging.info(f"Sucesso: {len(data)} registros obtidos no endpoint {endpoint}.")
        return data

    except requests.exceptions.RequestException as e:
        logging.error(f"Erro na requisição para a API ({url}): {e}")
        return []

def save_to_collection(db, data: list, collection_name: str, unique_keys: list) -> None:

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    if not data:
        logging.warning(f"Nenhum dado para salvar na collection '{collection_name}'. Processo ignorado.")
        return

    collection = db[collection_name]
    operacoes = []

    for doc in data:
        filtro = {key: doc.get(key) for key in unique_keys if doc.get(key) is not None}

        if len(filtro) != len(unique_keys):
            logging.debug(f"Documento ignorado por falta de chaves únicas {unique_keys}: {doc}")
            continue

        operacoes.append(
            UpdateOne(filtro, {"$set": doc}, upsert=True)
        )

    if operacoes:
        try:
            resultado = collection.bulk_write(operacoes)
            logging.info(
                f"Collection '{collection_name}' sincronizada: "
                f"{resultado.upserted_count} novos inseridos, "
                f"{resultado.modified_count} existentes atualizados."
            )
        except PyMongoError as e:
            logging.error(f"Erro ao persistir dados na collection '{collection_name}': {e}")

def main():
    db = conectar_mongodb()

    target_session_key = 9159
    target_meeting_key = 1219
    query_params = {"session_key": target_session_key}

    logging.info(f"--- Iniciando ETL para session_key={target_session_key} | meeting_key={target_meeting_key} ---")

    sessions_data = fetch_data("/sessions", query_params)
    save_to_collection(db, sessions_data, "sessions",
        unique_keys=["session_key"])

    drivers_data = fetch_data("/drivers", query_params)

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    save_to_collection(db, drivers_data, "drivers",
unique_keys=["session_key", "driver_number"])

    laps_data = fetch_data("/laps", query_params)
    save_to_collection(db, laps_data, "laps", unique_keys=["session_key",
"driver_number", "lap_number"])

    logging.info("--- Pipeline ETL finalizado com sucesso ---")

if __name__ == "__main__":
    main()
PS C:\Users\Adm\OneDrive\Documentos\vscodeigos\poo\P00> & C:\Users\Adm\AppData\Local\Programs\Python\
Python313\python.exe c:/Users/Adm/OneDrive/Documentos/vscodeigos/poo/P00/cartola.py/attdb/script.py
2026-09-27 12:27:48,486 - INFO - Iniciando conexão com o MongoDB...
2026-09-27 12:27:48,497 - INFO - Conectado com sucesso ao banco de dados: openf1_data
2026-09-27 12:27:48,497 - INFO - --- Iniciando ETL para session_key=9159 | meeting_key=1219 ---
2026-09-27 12:27:48,497 - INFO - Buscando /sessions | Parâmetros: {'session_key': 9159}...
2026-09-27 12:27:49,434 - INFO - Sucesso: 1 registros obtidos no endpoint /sessions.
2026-09-27 12:27:49,446 - INFO - Collection 'sessions' sincronizada: 1 novos inseridos, 0 existentes
atualizados.
2026-09-27 12:27:49,447 - INFO - Buscando /drivers | Parâmetros: {'session_key': 9159}...
2026-09-27 12:27:50,040 - INFO - Sucesso: 20 registros obtidos no endpoint /drivers.
2026-09-27 12:27:50,055 - INFO - Collection 'drivers' sincronizada: 20 novos inseridos, 0 existentes
atualizados.
2026-09-27 12:27:50,055 - INFO - Buscando /laps | Parâmetros: {'session_key': 9159}...
2026-09-27 12:27:50,797 - INFO - Sucesso: 475 registros obtidos no endpoint /laps.
2026-09-27 12:27:51,047 - INFO - Collection 'laps' sincronizada: 475 novos inseridos, 0 existentes a
tualizados.
2026-09-27 12:27:51,047 - INFO - --- Pipeline ETL finalizado com sucesso ---
PS C:\Users\Adm\OneDrive\Documentos\vscodeigos\poo\P00> █

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